Service Level Agreement between KU and Ankabut for Support Services and Expansion of High Performance Computing (HPC) and Cloud Services

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Approvals

By signing below, both parties agree to all terms and conditions mentioned in this document:

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Statement of Confidentiality

This agreement contains confidential information that is the property of Ankabut, the UAE Advanced Network for Research and Education. The recipient agrees to maintain such information confidential and not to copy nor disclose this information to any third party.

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1. Terms and Definitions

No	Term	Definition
5.1	Service	Means of delivering value for the customer by facilitating results the customer wants to achieve
5.2	Customer	Organization or part of an organization that receives a service or services
5.3	Service Level Agreement (SLA)	Documented agreement between Ankabut and a customer that identifies services and service targets
5.4	Incident	Unplanned interruption to a service, a reduction in the quality of a service or an event that has not yet impacted the service to the customer
5.5	Problem	Unknown cause of one or more incidents
5.6	Service Request	Request for information, advice, access to a service or a pre-approved change
5.7	Change	The addition, modification or removal of any service, service component or configuration item that could have an impact on services
5.8	Change Request	A formal proposal for a Change to be made to a service or service component (one or multiple Configuration Items)
5.9	Response Time	The time it takes to acknowledge a customer's incident in a non-automated way. It is measured from the time an incident record is created after contacting the service desk, until the time that the customer is advised that their incident has been received and is being addressed.
5.10	Resolution Time	The time it takes to resolve an incident. It is measured from the time an incident record is created after contacting the service desk, until the time that the customer accepts the resolution and closure.

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2. Executive Summary

Ankabut is the United Arab Emirates' Advanced National Research and Education Network (NREN) offering academic institutions connectivity to other education networks around the world. It also provides a wide range of Cloud Services and Managed IT Services that help organizations to collaborate, improve efficiency, optimize cost and reduce risk.

This document defines the Service Level Agreement (SLA) between **Ankabut** and **Khalifa University** of Science and Technology, referred in this document as **KU, Client or Customer.** The SLA covers:

1. Support Services for existing **High Performance Computing (HPC)** and **Cloud Advanced Services**
2. **Expansion of Cloud Infrastructure Services, HPC Cluster and Cloud Advanced Services.**

The HPC Services are based on a scale-out, flexible, and next generation HPC cluster platform to provide KU research community with appropriate infrastructure and facilities to boost research and development in the different fields that need computation-intensive applications.

The purpose of this SLA is to define the scope, roles and responsibilities, terms and conditions, and expected deliverables in order to provide ensure high quality services, efficient delivery and customer satisfaction.

3. Scope and Service Catalogue

3.1. Scope Summary

The following describes the main services provided as part of this agreement:

1. Hosting, Support and Subscription Services for Phase 1 and Phase 2 HPC and Cloud Advanced Services Systems
2. Cloud Infrastructure Services Expansion for 2023 and 2024 (*)
3. HPC, Cloud Advanced Services and Infrastructure Services for Phase 3 Expansion

(*) Note:

- Revisions will be agreed by the end of 2023 to accommodate additional resource needs for 2024. Such variation will be based on similar terms and conditions agreed in this SLA.

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3.2. Detailed Scope and Service Catalogue

#	Item
	Hosting, Support and Subscription Services for Phase 1 and Phase 2 HPC and Cloud Advanced Services Systems
1	Co-Location Hosting for 21 racks: 8 Rack : 18 for HPC, 2 for Cloud Advanced Services (MRS), 1 for Object Storage - hosting from Jan-2022
2	Hardware Support from 1-Jan-2022 and warranty extensions for Phase 1 from 15-Nov-2023 and Phase2 from 6-Nov-2024
3	Ankabut Cloud Management Subscription and support for Advanced Services for Big Data (MRS)
4	Ankabut Cloud Management Subscription and Support for Cloud Advanced Services for Object Storage
5	Monitoring and Alerting Services for HPC Resources
	Cloud Infrastructure Services Expansion for 2023 and 2024
6	Cloud Infrastructure as a Service (IaaS) for Research Computing Virtual Machines based on: - Reserved 490 vCPU - based on hyper-threating only (without any CPU over commitment) - Reserved RAM : 7440 GB
7	Managed Services for Research Computing VMs as per defined scope for 30 Windows OS VMs and 30 Linux OS VMs
8	Data Backup Services
	HPC, Cloud Advanced Services and Infrastructure Services Expansion Services for Phase 3 Expansion
9	Expansion of existing HPC Cluster with Compute and Storage Services (*): - Compute capacity = **3432 CPU-cores based on Intel CPU and 25,344GB RAM**. - Compute nodes will have 960GB SSD and Infiniband card-EDR 100Gb/s (Mellanox ConnectX-5 VPI) - Single port-QSFP28*1. - HPC Storage capacity: 800 TB
10	Expansion Cloud Advanced Services with: - Services for Big Data Analytics (MRS) with 11 Nodes - Services for 450TB Object Storage
11	Cloud Infrastructure as a Service (IaaS) for Research Computing Virtual Machines based on: - Reserved 392 vCPU - based on hyper-threating only (without any CPU over commitment) - Reserved RAM : 5952 GB - 400TB NAS Storage

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(*) Notes:

- As an indication, the total compute nodes will be **66** while using 2xIntel CPU 1288H V5, Intel Xeon Gold 6230R per node with 12x32G RAM
- HPC expansion will use existing shared infrastructure and software (Management Nodes, Login Nodes, Network Infrastructure, SLURM Job scheduler)
- **HPC Management Services.** Services will be with joint operations with KU research computing team for changes, upgrades, and the provision of the appropriate access and resources to KU and Ankabut Customers.

3.3. Managed Services for Virtual Machines

The Managed Services covers the System Administration Services for 60 VMs

Managed Services	Frequency
OS Patching service	Monthly for Critical patches
OS updates service	As required
Monitoring and alerting service	Daily
Server Resources Health check	Bi-weekly
Antivirus patching and update	Monthly
Backup services	If included in scope
Software installation for Windows servers	As required (*)
One-Time (deployment and configuration)	Yearly

(*) Note: It is assumed that 30 out of the 60VMs will be running Windows OS. Any software Installation for the remaining Linux systems will be done by the Research Computing Department team.

3.4. Data Backup Services

The data backup profile is described below:

Backup Profile

• Backup Schedule: Daily Incremental, Weekly Full.
• Backup Retention:
  ○ Daily Backups (in Days): 30.
  ○ Weekly Backups (in Weeks): 6.
  ○ Monthly Backups (in Months): 3.
  ○ Yearly Backups (in Years): 1.
• Years to Model: 5.
• Backup Copies: 1.

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Backup Services Scope	Quantity	Unit
Virtual Machine (VM) Backup License	60	VM
Bare Metal - Physical Hardware (PH) Backup License	10	PH
Backup Object Storage - OS for VMs	91	TB
Backup NAS Storage for HPC	13	TB
Backup Storage - Project data	236	TB
Backup Object Storage for RDISC VMs	3	TB
Backup Object Storage for RDISC Data	616	TB

3.5. Out of Scope

- Any service not included in the service catalog and scope
- Changes, enhancements, upgrades and development related to the items indicated in the service catalog.
- Software installations not related to the Service Catalog items.

3.6. Service Location

No.	Institute Name and Location
1	KU Main Campus

4. Roles & Responsibilities

4.1. Ankabut Roles & Responsibilities

- Providing high quality services as specified in the scope and related service catalog, and ensuring periodic review of the service performance.
- Meeting the performance levels and targets mentioned in this SLA.
- Providing regular reports and updates to the Client regarding the SLA adherence.
- Providing support and Service Management as per the best practices and international standards.

4.2. Client Roles & Responsibilities

- Provide resources to facilitate Ankabut and its subcontractor's staff performing the HPC and Cloud Advanced Services expansion and configuration.
- Coordinate with Ankabut for the HPC Management Platform and Cloud Advanced Services changes and updates related to KU.
- Manage HPC Cluster workloads and related rules for KU.
- Provide any scientific or technical support for KU HPC users.

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- Install, maintain and support all research software and tools required by KU users.
- Report any issue or service request related to Ankabut scope using the agreed communication channels and escalation levels (Section 8 of this document)

5. Governance Framework

Governance will be based on the overall contract between KU and Ankabut for Managed IT services

6. Service Quality

6.1. Service Support Hours

Contact	Working Hours
Service Desk (through Service Desk Portal, phone and email)	8x5 basis

6.2. Support Approach

- Ankabut to follow best practices and international standards related to IT Service Management (ITSM) through the following procedures:
  - Incident Management Procedure
  - Problem Management Procedure
  - Request Fulfilment Procedure for Managed Services
  - Change Management Procedure
- Customer can communicate with Ankabut Service Desk through the channels mentioned in Section 8 to report and log any issue or request.
- Ankabut will provide the customer with a reference number and will handle the request or resolve the issues as per the service levels mentioned in this document
- Ankabut shall keep the customer updated about the progress, and the Customer can escalate if required, as per the escalation levels in Section 9.

6.3. Support Levels and Targets

Support levels will be as per the contract and service level agreement between KU and Ankabut.

6.4. Service Maintenance

- Ankabut will perform regular service maintenance which might cause service interruptions.
- **Planned Outage:** For maintenance activities that require downtime/service interruption, Ankabut will send a notification to the Customer at least five business days before the

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planned implementation time. The notification will be sent via email, and will include a detailed description of the activity and the expected duration of the downtime/service interruption.

- **Unplanned Outage:** in case of any unplanned outage/service interruption, Ankabut will send a notification to the Customer via email, which include the details of the outage and expected resolution time. Another notification will be sent once the services are restored.

6.5. Service Reporting & Communications

No	Type (Report/Meeting)	Details	Frequency
1	Report	Tickets Workload, Resolution Progress and Service Level Adherence	Quarterly

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7. Service Rates

#	Item	Total for 2022	Total for 2023	Total for 2024
	Hosting, Support and Subscription Services for Phase 1 and Phase 2 HPC and Cloud Advanced Services Systems			
1	Co-Location Hosting for 21 racks			
2	Hardware Support from 1-Jan-2022 and warranty extensions for Phase 1 from 15-Nov-2023 and Phase2 from 6-Nov-2024			
3	Ankabut Cloud Management Subscription and support for Advanced Services for Big Data (MRS)			
4	Ankabut Cloud Management Subscription and Support for Cloud Advanced Services for Object Storage			
5	Monitoring and Alerting Services for HPC Resources			
	Sub-Total	**2,764,878**	**3,033,211**	**3,283,211**
	Cloud Infrastructure Services Expansion for 2023 and 2024			
6	Cloud Infrastructure as a Service (IaaS) for Research Computing Virtual Machines based on: - Reserved 490 vCPU - based on hyper-threating only (without any CPU over commitment) - Reserved RAM : 7440 GB	-	573,991	573,991
7	Managed Services for Research Computing VMs as per defined scope for 30 Windows OS VMs and 30 Linux OS VMs	-	363,038	363,038
8	Data Backup Services		695,161	695,161
	Sub-Total	**-**	**1,632,190**	**1,632,190**
	HPC, Cloud Advanced Services and Infrastructure Services Expansion Services for Phase 3 Expansion			
9	Expansion of existing HPC Cluster with Compute and Storage Services: - Compute capacity = 3432 CPU-cores based on Intel CPU and 25,344GB RAM. - Compute nodes have 960GB SSD and Infiniband card-EDR 100Gb/s (Mellanox ConnectX-5 VPI) - Single port-QSFP28*1. - HPC Storage capacity: 800 TB			
10	Expansion Cloud Advanced Services with: - 11 MRS Nodes - Big Data - 450TB Object Storage			

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11	Cloud Infrastructure as a Service (IaaS) for Research Computing Virtual Machines based on: - Reserved 392 vCPU - based on hyper-threating only (without any CPU over commitment) - Reserved RAM : 5952 GB - 400TB NAS Storage			
	Sub-Total	**7,855,290**	**3,927,645**	**3,927,645**
	TOTAL excluding VAT (AED)	**10,620,168**	**8,593,047**	**8,843,047**

Pricing Summary

#	Item	Total for 2022	Total for 2023	Total for 2024
1	Hosting, Support and Subscription Services for Phase 1 and Phase 2 HPC and Cloud Advanced Services Systems	2,764,878.00	3,033,211.33	3,283,211.33
2	Cloud Infrastructure Services Expansion for 2023 and 2024	-	1,632,190.29	1,632,190.29
3	HPC, Cloud Advanced Services and Infrastructure Services Expansion Services for Phase 3 Expansion	7,855,289.88	3,927,644.94	3,927,644.94
	TOTAL excluding VAT (AED)	**10,620,167.88**	**8,593,046.56**	**8,843,046.56**
	TOTAL excluding VAT for 3 Years (AED)	**28,056,261**		

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8. Contacts & Escalations

8.1. Ankabut Service Desk Contact Channels

There are four options available to contact Ankabut Service Desk

Office Location	Ankabut Offices, 1st Floor, Khalifa University Main Campus, Abu Dhabi
Phone	+971 2 6075999
Email	servicedesk@ankabut.ae
Web Portal	http://reach.ankabut.ac.ae

8.2. Escalations

Escalation Level	Person/Group
Level 1	Ankabut Service Desk
Level 2	Service Delivery and Support Manager
Level 3	Director of Technology Operations
Level 4	Ankabut CEO

9. Terms & Conditions

9.1. Duration

This SLA shall be valid with effect from the effective date and continues to be in force for three (3) years unless three (3) months' notice of termination is given by either party. The SLA may be renewed or extended on or before expiry date by mutual consent of both parties for such SLA.

9.2. Termination

The SLA shall remain valid unless three (3) months' notice is given by First Party to terminate the SLA or part of the SLA, if the Contractor's performance, response, behavior was found unsatisfactory or for any other reason opted by First Party, or upon three (3) months' notice given by Second Party for justifiable reasons that are acceptable to First Party. Also, either party may terminate the SLA with a minimum of three (3) months' notice

9.3. Variation

Amendments to any term, condition, or provisions of this SLA can only be effected by executing a Variation Order duly approved by Parties thereto. A Variation Order thus executed shall be deemed

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an integral part of this SLA. The Contractor shall strictly comply with all terms and conditions of this SLA and shall not carry any amendments to this SLA until formally authorized by executing a Variation Order duly approved by parties thereto.

9.4. Force Majeure

Failure or omission on the part of either Party to observe any of the terms and conditions hereinbefore stipulated shall not give rise to any claim or be deemed a breach of this SLA if such failure or omission arises from any cause reasonably beyond the control of either party such as fire, war, and natural disasters.

- Party claiming force majeure shall notify the other Party of such event within fifteen (15) days by email stating the estimated consequences. Onus to prove the occurrence of such events lies on the party claiming its invocation. If the force majeure lasts more than three (3) months the Client may terminate the SLA without affecting its rights adversely under this SLA.
- Any extension in the delivery period due to clauses of force majeure shall be by mutual agreed between Parties.

9.5. Payment Terms

- Payment to be made yearly in advance. The payment will be made within 30 days after receiving the original invoice.
- KU has the option to pay partial of the year or for more than one year's cost subject to mutual agreement by both parties.
- The prices provided in this SLA are valid for the entire SLA period as per Section 7 of the SLA.
- The SLA prices are in United Arab Emirates Dirhams and payments shall be made in such currency.
- In consideration of First Party payment of the SLA price, Second Party undertakes to duly carry out and deliver all the Services set forth in the scope of services detailed in this SLA hereto and carry out all other necessary Services and matters which can be implicitly understood in this SLA at the times, the manner and under the same terms and condition contained herein.
- Price of the work provided for in this SLA represents the full compensation payable to the Second Party against the Services, inclusive of all the costs and expenses for the due care, agency, taxes, customs, fees, charges, differences in the foreign exchange rates and any other burdens of any kind solely incurred by the Second Party.
- Prices are fixed and may be only increased upon the mutual consent of both parties.
- The period of time specified for determination of the due payment shall be with reference to the date at which First Party receives the correct invoice and any other relevant documents.
- Second Party has to present the original invoices marked with the SLA number.
- All Fees are exclusive of VAT or any additional taxes.

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